

Exam: Logic

1. Propositional and First-Order Logic

1. Determine whether the following statements are **tautologies**, **contradictions**, or **contingent**:
 - a) $P \vee \neg P$
 - b) $(P \vee Q) \wedge (\neg P \vee \neg Q)$
 - c) $(P \rightarrow Q) \vee (Q \rightarrow P)$
 - d) $(P \wedge Q) \rightarrow P$
 2. Construct the **truth table** for the following expressions:
 - a) $(P \vee Q) \rightarrow R$
 - b) $(P \wedge Q) \vee (\neg P \wedge \neg Q)$
 - c) $\neg(P \vee Q) \equiv \neg P \wedge \neg Q$
 3. Translate the following statements into **propositional logic**:
 - a) “If it is raining, then the ground is wet.”
 - b) “Either Alice or Bob will win the competition, but not both.”
 - c) “If the alarm is not working, then either there is a power failure or the battery is dead.”
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2. Formalization of Arguments

4. Determine whether the following arguments are **valid** using **truth tables** or **logical inference rules**:
 - a) **Premises:**
 - $P \rightarrow Q$
 - $Q \rightarrow R$
 - P**Conclusion:** R
 - b) **Premises:**
 - $P \vee Q$
 - $\neg P$**Conclusion:** Q
 - c) **Premises:**

- $(P \rightarrow Q) \wedge (Q \rightarrow R)$

- $\neg R$

Conclusion: $\neg P$

5. Use **natural deduction** to prove the following statements:

a) $P \vee (Q \wedge R) \vdash (P \vee Q) \wedge (P \vee R)$

b) $(P \rightarrow Q) \wedge (R \rightarrow S), P \vee R \vdash Q \vee S$

3. Applications in Mathematics and Computation

6. **Set Theory Application:** Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$, $A = \{2, 4, 6, 8\}$, and $B = \{1, 2, 3, 4\}$. Compute:

a) $A \cup B$

b) $A \cap B$

c) A^c (the complement of A in U)

d) $(A \cup B)^c$

7. **Boolean Algebra and Circuits:** Simplify the following Boolean expressions using **Boolean algebra laws**:

a) $(A \vee B) \wedge (\neg A \vee B)$

b) $(A \wedge B) \vee (\neg A \wedge B)$

c) $A \vee (A \wedge B)$

8. **Logic Programming (Prolog):** Given the facts:

```
parent(alice, bob).
parent(bob, charlie).
parent(charlie, david).
```

a) Write a **recursive rule** for the “ancestor” relation.

b) Query whether **Alice is an ancestor of David** using your rule.